Haemorrhagic Cystitis and Renal Dysfunction

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51.1 Haemorrhagic Cystitis

51.1.1 Introduction

Haemorrhagic cystitis (HC) is a frequent complication after haematopoietic cell transplantation (HCT). According to the time of occurrence after HCT, HC is defined as early onset and late onset. Early-onset HC occurs typically during or within 48 h after the end of the conditioning regimen, and it is the result of a direct toxic effect of drug metabolites and radiotherapy on the bladder mucosa. Late-onset HC usually starts around the time of neutrophil engraftment (weeks 2–4) up to the second to third month after HCT (Hirsch and Pergam 2016). Haematuria, with or without symptoms, is classified into four categories: microscopic (grade 1), macroscopic (grade 2), macroscopic with clots (grade 3) and macroscopic with clots and renal failure secondary to urinary tract obstruction (Droller et al. 1982; Bedi et al. 1995). HC is associated with haematuria >grade 2.

51.1.2 Risk Factors

The presence of pro-haemorrhagic abnormalities of coagulation, severe thrombocytopenia and mucosal inflammation are predisposing factors for any type of HC.

The main risk factor for late-onset HC is infection by polyomavirus BK (BKPyV), whereas other viruses such as *adenovirus*, *cytomegalovirus* and *JC polyomavirus* have been rarely implicated. BKPyV is a common cause of asymptomatic or mild flu-like infection during early infancy and childhood, and more than 90% of adults are seropositive for BKPyV. The route of transmission is by droplets or contact with oral saliva and respiratory tract secretions. The virus persists latently in renal tubular epithelial and urothelial cell and can replicate as the host virus-specific T-cell response is lost or severely weakened. Mild to moderate asymptomatic BKPyV viruria is seen in 5–10% of healthy individuals, especially the oldest and the pregnant women, whereas high-load BKPyV viruria is detected in 50–60% of patients who underwent allogeneic HCT due to the severe immunosuppression condition.

BKPyV viruria develops in more than 50% of allogeneic HCT, but overt HC occurs in about 20% of patients because several patient- or transplant-related factors affect its occurrence: the type of graft (CB and PB > BM); the type of donor (Haplo > URD > MRD); the type of conditioning regimen (MAC > RIC); the use in the conditioning regimen of ATG, CY, or BU; the occurrence of acute GVHD grade 2–4; and, among the paediatric patients, a recipient age >7 years.

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51.1.3 Pathogenesis

The cause of post-HCT HC is multifactorial and includes the combined effects of the extensive viral cytopathic damage of the bladder mucosa, the chemical or actinic damage induced by the conditioning regimen and the immune donor-derived alloreactivity targeting bladder mucosa (Cesaro et al. 2018). In patients receiving allogeneic HCT, BKPyV viruria >10^7 genomic copies/mL and BKPyV viremia >10^3 genomic copies/mL are predictive factors for BKPyV-HC (Cesaro et al. 2015).

51.1.4 Diagnosis

The diagnosis of BKPyV-HC is defined by the presence of macrohaematuria (>grade 2), of clinical symptoms/signs of cystitis (dysuria, increased urinary frequency, lower abdominal pain) and of high-load BKPyV viruria (>10^7) (Cesaro et al. 2018). Two-thirds of patients have also BKPyV viremia (>10^3) (Erard et al. 2005; Cesaro et al. 2015). Other infectious and non-infectious causes of HC must be excluded. Instrumental examination such as ultrasound or CT scan shows signs of bladder inflammation (oedema and thickening of bladder wall) and the presence of intra-bladder clots or obstruction of urinary tract.

The reduction of both BKPyV viruria and BKPyV viremia has been correlated with clinical recovery from HC. The monitoring of BKPyV viruria and viremia of HCT patients is not recommended since the type of pre-emptive intervention is not established.

51.1.5 Prophylaxis

Effective preventive measures are possible only for early-onset HC in patients who receive high dose of CY as part of the conditioning regimen or GVHD prophylaxis (HCT with PTCy). These are based on hyperhydration and the administration of continuous intravenous infusion of mesna, which reduce the exposure of the bladder mucosa to acrolein and other toxic catabolites.

The bladder irrigation through a two- or three-way urinary catheter is no more effective than hyperhydration, and considering the invasiveness and patient discomfort, its use is not recommended.

In the late-onset HC, BKPyV replication has a key role in exacerbating the damage of bladder mucosa through its cytopathic effect and in inducing the donor immune alloreactivity targeting the bladder mucosa. Although fluoroquinolones inhibit in vitro BKPyV, the efficacy in vivo is weak and do not affect significantly the incidence of HC. Considering the risk of inducing bacterial resistance and tendon and joint damages in children, fluoroquinolones are not recommended for HC prophylaxis.

51.1.6 Treatment

Cidofovir is a nucleotide analogue inhibiting several DNA viruses including BKPyV. Its long half-life (15–65 h) allows the administration every 1 or 2 weeks. Given the significant risk of tubular nephrotoxicity, cidofovir has been used only for therapeutic purposes (Cesaro et al. 2009). The nephrotoxicity can be limited by saline hydration and by the use of probenecid that inhibits the capture and transport of cidofovir into the renal tubular epithelial cells. There is no agreement on the optimal dose, modality of administration and frequency of administration. One scheme is the intravenous administration of cidofovir, 3–5 mg/kg body weight, weekly or fortnightly, together with probenecid to prevent nephrotoxicity. Mild to moderate increase in serum creatinine is observed in 18% of the patients. Another scheme is the administration of low-dose intravenous cidofovir, 0.5–1.5 mg/kg body weight, 1–3 times a week, without probenecid (Ganguly et al. 2010). Mild to moderate renal toxicity is reported in 20% of patients. Alternatively, cidofovir can be administered intravesically to reduce the risk of nephrotoxicity, at the dose of 5 mg/kg/body weight/week and left in situ for 1–2 h after clamping the vesical catheter, the response rate being about 50% (Bridges et al. 2006).

Some results have been reported with leflunomide, an antimetabolite drug with immunomodulatory and antiviral activity. Anecdotal use of other agents (vidarabine, oral levofloxacin, FXIII concentrate, intravesical sodium hyaluronate and oestrogens) is reported in the literature (Cesaro et al. 2018).

The recovery from HC, whatever the cause, can benefit from treatment aiming to repair and regenerate the urothelial mucosa (hyperbaric oxygen therapy) and to stop bleeding (topical application of fibrin glue or platelet-rich plasma). The main drawback of hyperbaric oxygen is the limited availability, the requirements for dedicated hyperbaric room facilities, the risk of ear barotrauma or pressure intolerance and claustrophobia episodes during the procedure (Zama et al. 2013; Cesaro et al. 2018). Cystoscopic application of fibrin glue to the damaged bleeding bladder mucosa has been associated with a response rate of 83%, with most of cases resolved with one or two applications (Tirindelli et al. 2014).

The mesenchymal stromal cells (MSC) and the adoptive immunotherapy represent the most recent innovative treatment experimented for HC. MSC have the potential to stimulate the tissue repairing process and exercise an immune modulatory and anti-inflammatory effect. The use of third-party MSC infusion into seven patients with BKPyV-HC obtained the resolution of haematuria in five patients (Ringden et al. 2007). This approach needs to be validated further to assess the feasibility and also the safety of MSC.

Adoptive transfer of donor-derived virus-specific T cells (VSTs) has shown efficacy for the treatment of several viral infections although their use on a larger scale is limited by the costs, the complexity of manufacturing, the time needed to obtain the final cell product that is not suitable for the urgent treatment and the prompt availability of a seropositive donor. The use of banked VSTs obtained by a third-party healthy seropositive donor, cryopreserved and used as the patient develops a viral infection refractory to antiviral treatment represents a promising development (Olson et al. 2021). In phase II trials, the use of VSTs directed against BKPyV obtained an overall response rate of 81–92%. The infusions of VSTs resulted safe with no cases of moderate–severe GVHD. Importantly, the VSTs expanded in vivo and the functionality persisted for up to 12 weeks (Tzannou et al. 2017). These results are encouraging and they need the confirmation by prospective larger studies.

51.2 Renal Dysfunction

Acute kidney injury (AKI) occurs in 27–66% of patients who underwent allo-HCT mainly within the first 100 days. The incidence of AKI is lower in autologous than in allogeneic HCT due to several reasons: rapid engraftment, lower incidence of infectious complications, absence of GVHD and its inflammatory cytokines, absence of CMV infection, lower frequency of severe diarrhoea and dehydration and less drug-induced nephrotoxicity (Lopes et al. 2016; Raina et al. 2017).

The diseases associated with AKI act at different renal levels: prerenal (sepsis, engraftment syndrome, SOS/VOD), renal glomerular (transplant-associated microangiopathy), renal tubular (acute tubular necrosis due to dehydration, sepsis, shock, engraftment syndrome, intratubular obstruction due to drugs, or tumour lysis syndrome), renal interstitial (acute GVHD, viral infection by BKPvyV or ADV) and post-renal (obstruction by BKPyV or adenovirus cystitis, retroperitoneal fibrosis, lymphadenopathy).

General risk factors for AKI are pre-HCT diabetes, hypertension and renal impairment; the use of nephrotoxic drugs for the conditioning regimen (ifosfamide, CY, carboplatin, cisplatin), for the treatment of GVHD (MTX, cyclosporine, tacrolimus), for the treatment of infections (liposomal amphotericin B, aminoglycosides, vancomycin) and for the treatment of other severe organ damages that require ICU admission and mechanical ventilation (Hingorani 2016).

Clinically, the severity of AKI is defined by serum creatinine (SCr) and urine output (UO) that permits the identification of three groups: patient at risk of AKI (SCr increase of 1.5–2× and

UO <0.5 mL/kg/h for >6 h); patient with kidney injury (SCr increase of 2–3× and UO <0.5 mL/kg/h for >12 h); and patient with kidney failure (SCr increase of >3× and UO <0.3 mL/kg/h for >24 h or anuria >12 h, or initiation of replacement therapy) (Lopes et al. 2016).

AKI represents a risk factor for the development of chronic kidney disease on the medium–long-term period, especially if the acute damage is not completely resolved and proteinuria and hypertension persist and for the increase of non-relapse and overall mortality (Shingai et al. 2015).

Key Points

	Early-onset HC	Late-onset HC	Comments
Incidence	<3%	7–25%	Early-onset HC is nowadays rare
Pathogenesis	Chemical (drugs) or actinic damage of bladder mucosa	BK virus infection Adenovirus infection Donor alloreactivity	
Diagnosis	Macrohaematuria with dysuria, increased urinary frequency, low abdominal pain	Macrohaematuria with dysuria, increased urinary frequency, low abdominal pain, high load of BK viruria and/or viremia	Signs of bladder inflammation at ultrasound examination
Prevention	Hyperhydration, mesna (if chemo with Cy), forced diuresis	Hyperhydration, forced diuresis	Fluoroquinolones not recommended
Therapy	Hyperhydration Forced diuresis Hyperbaric O_2 therapy (HOT) Application of fibrin glue by cystoscopy	Hyperhydration, forced diuresis IV (or intravesical) cidofovir HOT Application of fibrin glue or platelet-rich plasma by cystoscopy	No agreement On dose and route of Cidofovir administration Limited evidence/experience for HOT, fibrin glue, platelet-rich plasma and other treatments
Experimental	/	Virus-specific T cells (BK virus, Adenovirus) Mesenchymal cells	

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